

Exercise 2-4

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Exercise 2-Visualizing key characteristics

```
##Loading the required packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2     4.0.3      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr       1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at C:/Users/userx/Desktop/Assignments/week 4
```

```
library(rio)
```

```
##create the required folders in the working directory
dir.create(here("data"), showWarnings = FALSE)
dir.create(here("scripts"), showWarnings = FALSE)
dir.create(here("figures"), showWarnings = FALSE)
dir.create(here("reports"), showWarnings = FALSE)
```

```
# Install/load packages as needed
pacman::p_load(here, rio)
```

```
# Ensure the repository's data folder exists
```

```
dir.create(here("data"), showWarnings = FALSE, recursive = TRUE)

# Copy the source files into the repository's data folder
file.copy(
  from = "C:/Users/userx/Downloads/cleaned_NHANES (1).csv",
  to = here("data", "cleaned_nhanes.csv"),
  overwrite = TRUE
)
```

```
## [1] TRUE
```

```
file.copy(
  from = "C:/Users/userx/Downloads/diet (1).csv",
  to = here("data", "diet (1).csv"),
  overwrite = TRUE
)
```

```
## [1] TRUE
```

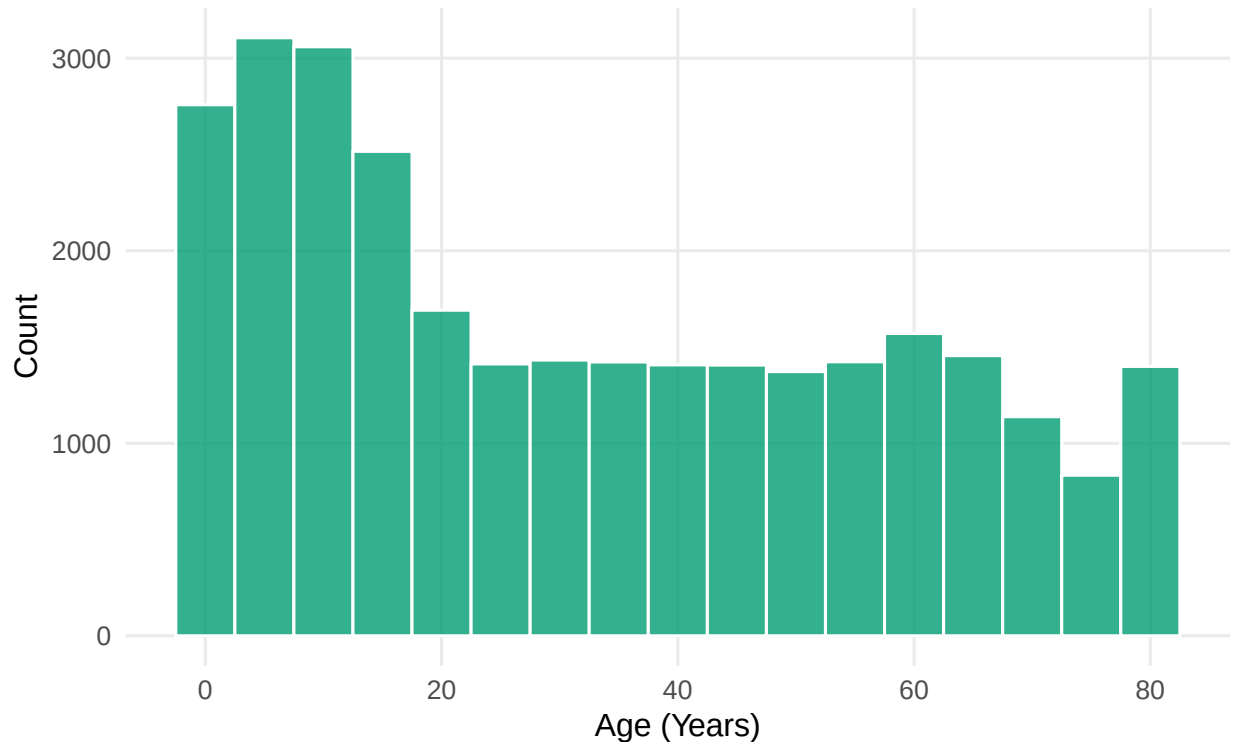
```
# Import the files after copying them
nhanes_data <- rio::import(here("data", "cleaned_nhanes.csv"))
diet_data <- rio::import(here("data", "diet (1).csv"))
```

```
##Using ggplot() to create appropriate visualisations
view(nhanes_data$age)
visual_age <- p_age <- ggplot(nhanes_data, aes(x = age)) +
  geom_histogram(binwidth = 5, fill = "#009E73", color = "white", alpha = 0.8) +
  labs(
    title = "Distribution of Age in the NHANES Dataset",
    subtitle = "Each bin represents a 5-year interval",
    x = "Age (Years)",
    y = "Count"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank()
  )

visual_age
```

Distribution of Age in the NHANES Dataset

Each bin represents a 5-year interval



```
##checking for missingness
sum(is.na(nhanes_data$age))
```

```
## [1] 0
```

```
sum(nhanes_data$age)
```

```
## [1] 956149
```

##Distribution of age For the variable age, the distribution shows a right-skewed pattern. The age of the patients range from 0 to 80 years with a median age of 28ears and average age of 32.5 years. There are no missing vales in the variable age.

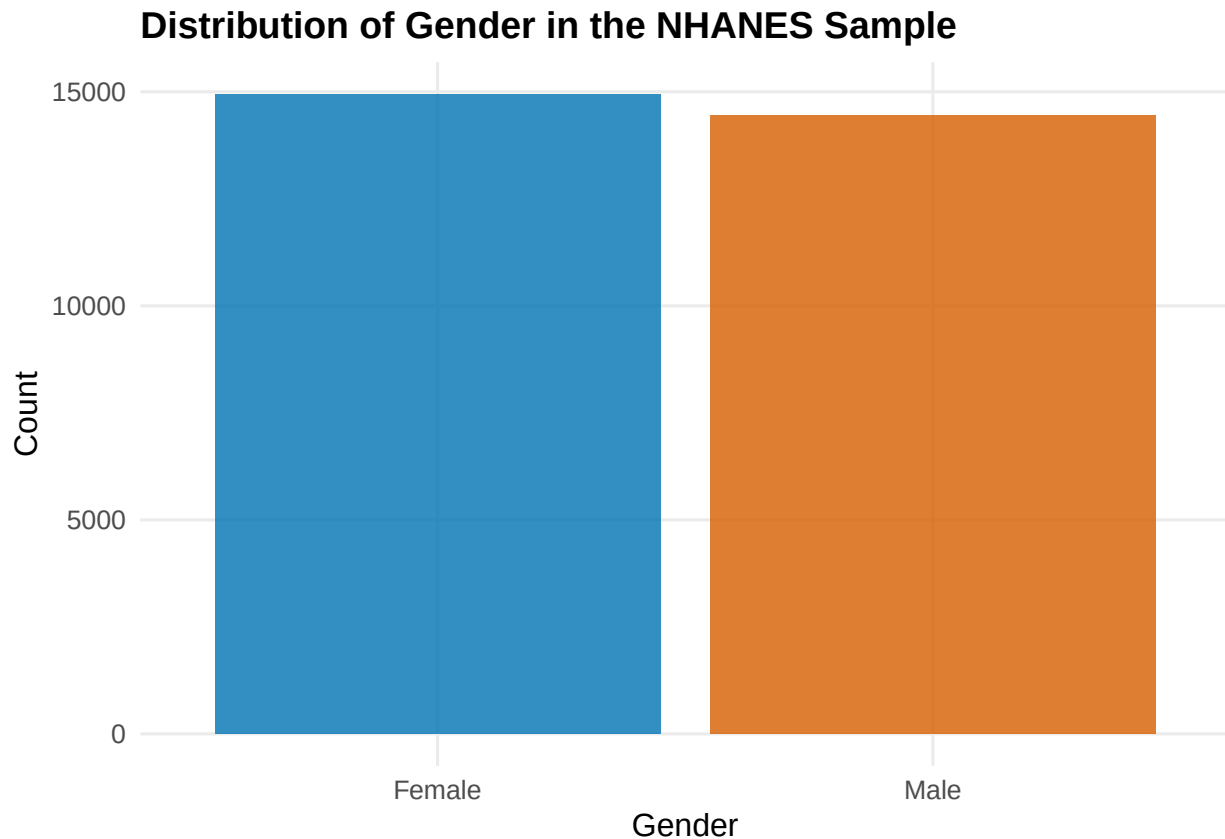
```
visual_gender <- ggplot(nhanes_data, aes(x = gender, fill = gender)) +
  geom_bar(alpha = 0.8) +
  scale_fill_manual(values = c("#0072B2", "#D55E00")) +
  labs(
    title = "Distribution of Gender in the NHANES Sample",
    x = "Gender",
    y = "Count",
    fill = "Gender"
  ) +
  theme_minimal(base_size = 12) +
  theme(
```

```

plot.title = element_text(face = "bold", size = 14),
legend.position = "none",
panel.grid.minor = element_blank()
)

```

visual_gender



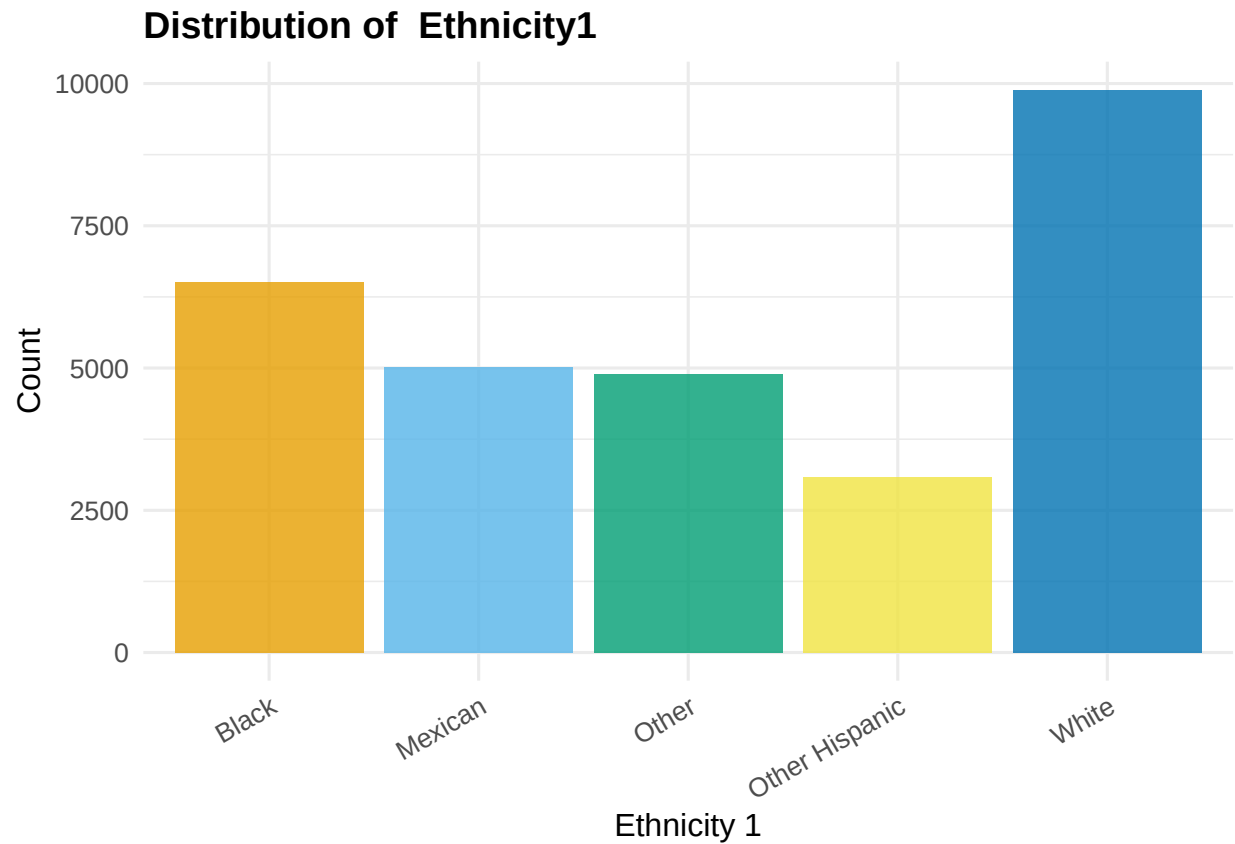
##Distribution of Gender Among the patients, (14,948)50.8% are female and 14,452(49.2%) and there are no missing values in the variable gender

```

##Distribution of Ethnicity1
visual_ethnicity_1 <- ggplot(nhanes_data, aes(x = ethnicity_1, fill = ethnicity_1)) +
  geom_bar(alpha = 0.8) +
  scale_fill_manual(values = okabe_ito_palette) +
  labs(
    title = "Distribution of Ethnicity1",
    x = "Ethnicity 1",
    y = "Count",
    fill = "Ethnicity1"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    axis.text.x = element_text(angle = 30, hjust = 1),
    legend.position = "none"
  )

```

```
visual_ethnicity_1
```



```
##checking for missingness  
sum(is.na(nhanes_data$ethnicity_1))
```

```
## [1] 0
```

```
table(nhanes_data$ethnicity_1)
```

```
##  
##      Black      Mexican      Other Other Hispanic      White  
##      6511      5018      4893      3088      9890
```

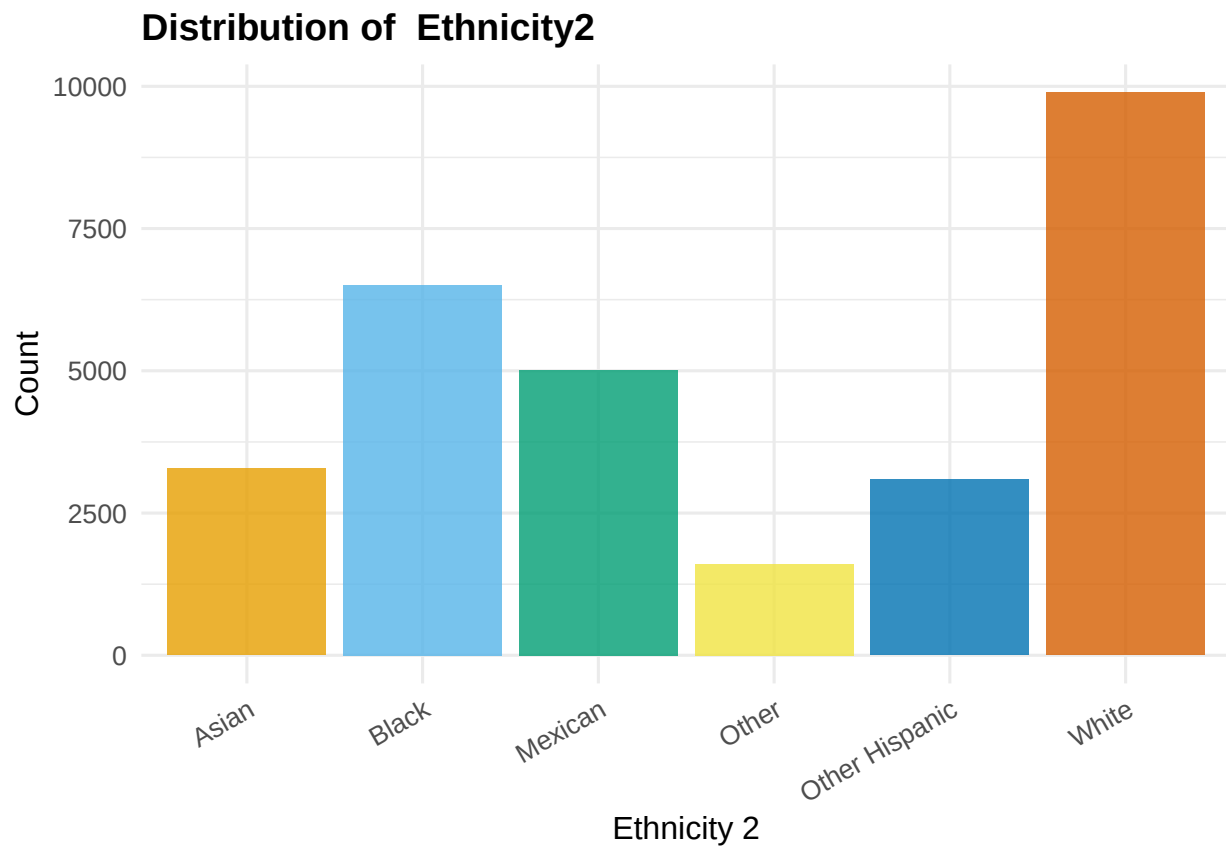
##Distribution of first category of ethnicity1 The total number of participants is 29,400, of these the majority are white 9,890 (33.6%) followed by the black participants 6,511 (22.2%). The lowest category of participants is hispanic 3,088 (10.5%). This category of ethnicity has no missing values.

```
visual_ethnicity_2 <- ggplot(nhanes_data, aes(x = ethnicity_2, fill = ethnicity_2)) +  
  geom_bar(alpha = 0.8) +  
  scale_fill_manual(values = okabe_ito_palette) +  
  labs(  
    title = "Distribution of Ethnicity2",  
    x = "Ethnicity 2",
```

```

  y = "Count",
  fill = "Ethnicity2"
) +
theme_minimal(base_size = 12) +
theme(
  plot.title = element_text(face = "bold", size = 14),
  axis.text.x = element_text(angle = 30, hjust = 1),
  legend.position = "none"
)
visual_ethnicity_2

```



```

##checking for missingness
ethnicity2_table <- table(nhanes_data$ethnicity_2, useNA = "ifany")

data.frame(
  ethin = names(ethnicity2_table),
  Frequency = as.vector(ethnicity2_table),
  Percentage = round(as.vector(prop.table(ethnicity2_table) * 100), 1)
)

```

```

##          ethin Frequency Percentage
## 1         Asian      3284         11.2
## 2         Black      6511         22.1
## 3         Mexican      5018         17.1

```

```
## 4      Other      1609      5.5
## 5 Other Hispanic  3088     10.5
## 6      White     9890     33.6
```

```
summary(nhanes_data$age)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.00   10.00   28.00   32.52   54.00   80.00
```

```
table(nhanes_data$gender)
```

```
##
## Female   Male
##  14948   14452
```

```
table(nhanes_data$ethnicity_1)
```

```
##
##      Black      Mexican      Other Other Hispanic      White
##      6511      5018      4893      3088      9890
```

```
table(nhanes_data$ethnicity_2)
```

```
##
##      Asian      Black      Mexican      Other Other Hispanic
##      3284      6511      5018      1609      3088
##      White
##      9890
```

```
##Using "ggsave" to save all plots in the figure subfolder in the working directory
```

```
##saving age
```

```
ggsave(
  filename = "age_distribution_plot.png",
  plot = visual_age,
  path = here("figures"),
  width = 8,
  height = 6
)
```

```
##saving gender
```

```
ggsave(
  filename = "gender_distribution_plot.png",
  plot = visual_gender,
  path = here("figures"),
  width = 8,
  height = 6
)
```

```
##saving ethnicity1
```

```
ggsave(
  filename = "Ethnicity1_distribution_plot.png",
  plot = visual_ethnicity_1,
```

```

    path = here("figures"),
    width = 8,
    height = 6
  )
  ##saving ethnicity2
  ggsave(
    filename = "Ethnicity2_distribution_plot.png",
    plot = visual_ethnicity_2,
    path = here("figures"),
    width = 8,
    height = 6
  )

```

```

##choosing from the two categories of ethnicity
##structure of the two categories
str(nhanes_data$ethnicity_1)

```

```
## chr [1:29400] "Black" "White" "White" "White" "White" "Mexican" "White" ...
```

```
str(nhanes_data$ethnicity_2)
```

```
## chr [1:29400] "Black" "White" "White" "White" "White" "Mexican" "White" ...
```

```

##tabulating the categories
table(nhanes_data$ethnicity_1)

```

```
##
##      Black      Mexican      Other Other Hispanic      White
##      6511      5018      4893      3088      9890
```

```
table(nhanes_data$ethnicity_2)
```

```
##
##      Asian      Black      Mexican      Other Other Hispanic
##      3284      6511      5018      1609      3088
##      White
##      9890
```

```

##checking for missingness
sum(is.na(nhanes_data$ethnicity_1))

```

```
## [1] 0
```

```
sum(is.na(nhanes_data$ethnicity_2))
```

```
## [1] 0
```

##Decision Ater analysing the two categories, i decided to keep Ethnicity2 category and drop category 1. Whereas both categories have an equal number of observations (29,400), ethnicity1 has 5 categories while ethnicity has 6 categories. Ethnicity2 adds on the “asian” category which makes it more representative than category 1.

```
##keeping Ethnicity
nhanes_data <- nhanes_data %>%
  select(-ethnicity_1)
names(nhanes_data)
```

```
## [1] "V1"          "patient_id"    "bp_comment"    "systolic_bp_1"
## [5] "diastolic_bp_1" "systolic_bp_2" "diastolic_bp_2" "systolic_bp_3"
## [9] "diastolic_bp_3" "systolic_bp_4" "diastolic_bp_4" "wave_id"
## [13] "gender"        "age"           "ethnicity_2"    "education"
## [17] "marital_status" "family_size"    "family_income"  "age_cat"
```

=====

Exercise 3-Improving ggplot figure

=====

##points that need improvement in your colleague's figure and how i will improve them. In my colleague's plot, there's no title, i will create a plot and give a title "Participant's Weight Change from Baseline weight". This helps the audience the the objective of the plot at first glance.

The objective of the study is to show how each participants weight had changed over time from the baseline weight. My colleague's plot doesn't show baseline weight. I will create a baseline weight (at 0) and show how each participant deviates from it. Those above it have had a weight gain and those below it have had a weight loss.

```
## To determine how each participant's weight has changed over time in comparison to the baseline weight
diet_data <- import(here("data", "diet (1).csv"))
```

```
change_in_weight <- diet_data %>%
  group_by(Participant) %>%
  arrange(Week) %>%
  mutate(
    baseline_weight = Weight[Week == 0][1],
    change_from_baseline = Weight - baseline_weight
  ) %>%
  ungroup()

head(change_in_weight)
```

```
## # A tibble: 6 x 5
##   Week Participant Weight baseline_weight change_from_baseline
##   <int>         <int>   <dbl>         <dbl>              <dbl>
## 1     0             1   95.9           95.9                0
## 2     0             2   95.8           95.8                0
## 3     0             3   69.5           69.5                0
## 4     0             4   91.4           91.4                0
## 5     0             5   83.8           83.8                0
## 6     0             6   80.2           80.2                0
```

```
change_in_weight
```

```
## # A tibble: 340 x 5
##   Week Participant Weight baseline_weight change_from_baseline
##   <int>         <int> <dbl>         <dbl>             <dbl>
## 1     0           1  95.9           95.9               0
## 2     0           2  95.8           95.8               0
## 3     0           3  69.5           69.5               0
## 4     0           4  91.4           91.4               0
## 5     0           5  83.8           83.8               0
## 6     0           6  80.2           80.2               0
## 7     0           7  88.3           88.3               0
## 8     0           8  64.3           64.3               0
## 9     0           9  84.6           84.6               0
## 10    0          10  86.9           86.9               0
## # i 330 more rows
```

From the results, 0 means there was no change in weight after following the diet, negative means there was a loss in weight and a positive means there was a gain in weight.

```
##Plotting the right spaghetti plot to show each participants weight has changed
library(RColorBrewer) # Required for ColorBrewer palettes
```

```
plot_data <- change_in_weight %>%
  group_by(Participant) %>%
  arrange(Week, .by_group = TRUE) %>%
  mutate(
    baseline_weight = first(Weight[Week == 0]),
    change_from_baseline = Weight - baseline_weight
  ) %>%
  ungroup()

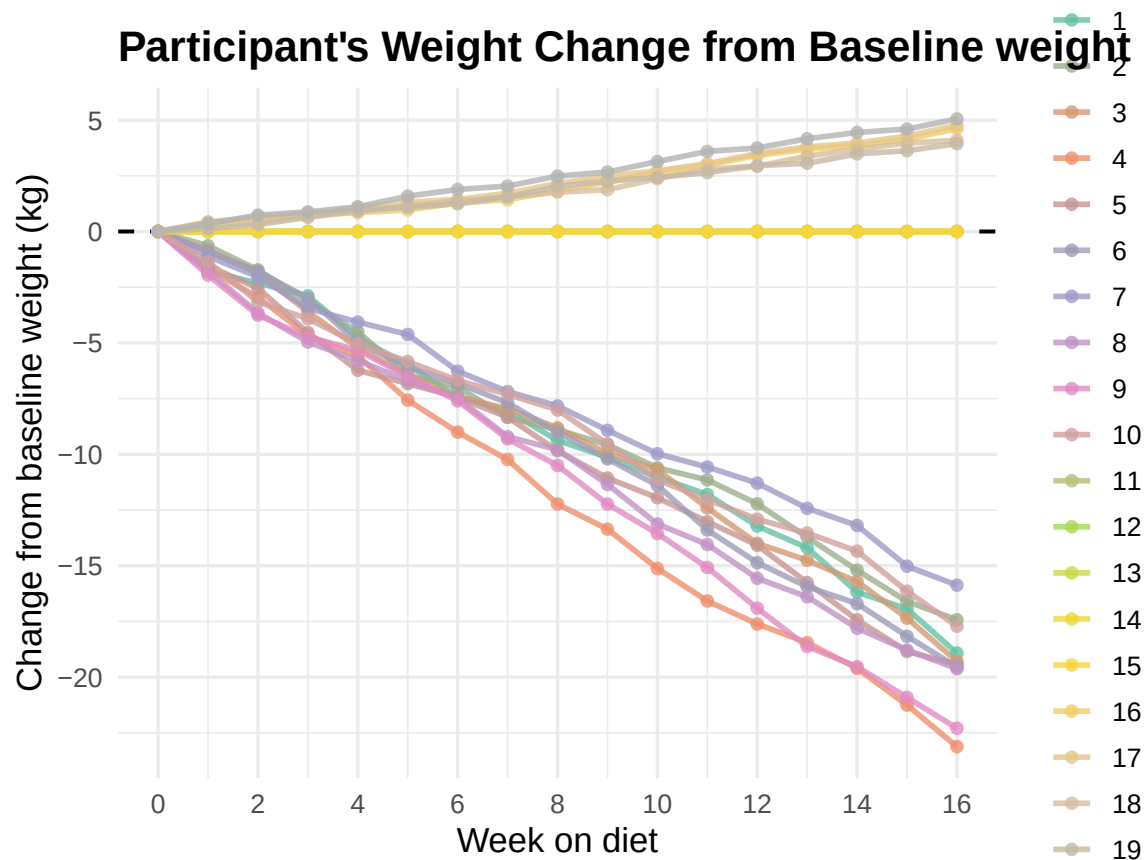
# 1. Generate a color palette using RColorBrewer
num_participants <- n_distinct(plot_data$Participant)
my_colors <- colorRampPalette(brewer.pal(8, "Set2"))(num_participants)

# 2. Build the plot with individual distinct colors
ggplot(
  plot_data,
  aes(
    x = Week,
    y = change_from_baseline,
    group = Participant,
    color = factor(Participant) # Map a unique color per participant
  )
) +
  # Reference line at zero (Baseline)
  geom_hline(
    yintercept = 0,
    linetype = "dashed",
    color = "black",
    linewidth = 0.7
  ) +
```

```

# Individual trajectories
geom_line(
  linewidth = 1,
  alpha = 0.8
) +
geom_point(
  size = 1.6,
  alpha = 0.8
) +
# Apply the RColorBrewer custom palette
scale_color_manual(values = my_colors) +
# X-axis formatting
scale_x_continuous(
  breaks = seq(0, 16, by = 2)
) +
# Professional labels and titles
labs(
  title = "Participant's Weight Change from Baseline weight",
  x = "Week on diet",
  y = "Change from baseline weight (kg)",
  color = "Participant"
) +
# Theme styling
theme_minimal(base_size = 13) +
theme(
  plot.title = element_text(face = "bold"),
  legend.position = "right" # Set to "none" if the legend is too large for your layout
)

```



Exercise 4-Exploring relationships through visualizations

##The objective of this part is to explore relationships between age, gender, and systolic blood pressure
##(SBP) using effective visual storytelling

```
names(nhanes_data)
```

```
## [1] "V1"          "patient_id"   "bp_comment"   "systolic_bp_1"
## [5] "diastolic_bp_1" "systolic_bp_2" "diastolic_bp_2" "systolic_bp_3"
## [9] "diastolic_bp_3" "systolic_bp_4" "diastolic_bp_4" "wave_id"
## [13] "gender"        "age"          "ethnicity_2"   "education"
## [17] "marital_status" "family_size"   "family_income" "age_cat"
```

```
view(nhanes_data)
```

```
##Averaging the systolic blood pressure variables
```

```
names(nhanes_data)
```

```
## [1] "V1"          "patient_id"    "bp_comment"    "systolic_bp_1"
## [5] "diastolic_bp_1" "systolic_bp_2" "diastolic_bp_2" "systolic_bp_3"
## [9] "diastolic_bp_3" "systolic_bp_4" "diastolic_bp_4" "wave_id"
## [13] "gender"        "age"           "ethnicity_2"   "education"
## [17] "marital_status" "family_size"   "family_income" "age_cat"
```

```
nhanes_data <- nhanes_data %>%
  mutate(
    avg_sbp = rowMeans(
      select(., systolic_bp_1, systolic_bp_2, systolic_bp_3, systolic_bp_4),
      na.rm = TRUE
    ),
    avg_sbp = ifelse(is.nan(avg_sbp), NA, avg_sbp)
  )
##Create a new categorical variable that classifies individuals based on their average SBP
nhanes_data <- nhanes_data %>%
  mutate(
    sbp_category = case_when(
      avg_sbp < 120 ~ "Normal",
      avg_sbp >= 120 & avg_sbp < 130 ~ "Elevated",
      avg_sbp >= 130 & avg_sbp < 140 ~ "Stage 1 hypertension",
      avg_sbp >= 140 ~ "Stage 2 hypertension",
      TRUE ~ NA_character_
    ),
    sbp_category = factor(
      sbp_category,
      levels = c("Normal", "Elevated", "Stage 1 hypertension", "Stage 2 hypertension")
    )
  )
names(nhanes_data)
```

```
## [1] "V1"          "patient_id"    "bp_comment"    "systolic_bp_1"
## [5] "diastolic_bp_1" "systolic_bp_2" "diastolic_bp_2" "systolic_bp_3"
## [9] "diastolic_bp_3" "systolic_bp_4" "diastolic_bp_4" "wave_id"
## [13] "gender"        "age"           "ethnicity_2"   "education"
## [17] "marital_status" "family_size"   "family_income" "age_cat"
## [21] "avg_sbp"       "sbp_category"
```

```
view(nhanes_data)
##Use the filter() function to create a new datasets that includes only individuals who have
#non-missing values for age, gender, and average SBP.
nhanes_clean <- nhanes_data %>%
  filter(!is.na(age_cat) & !is.na(gender) & !is.na(avg_sbp))
view(nhanes_clean)
names(nhanes_clean)
```

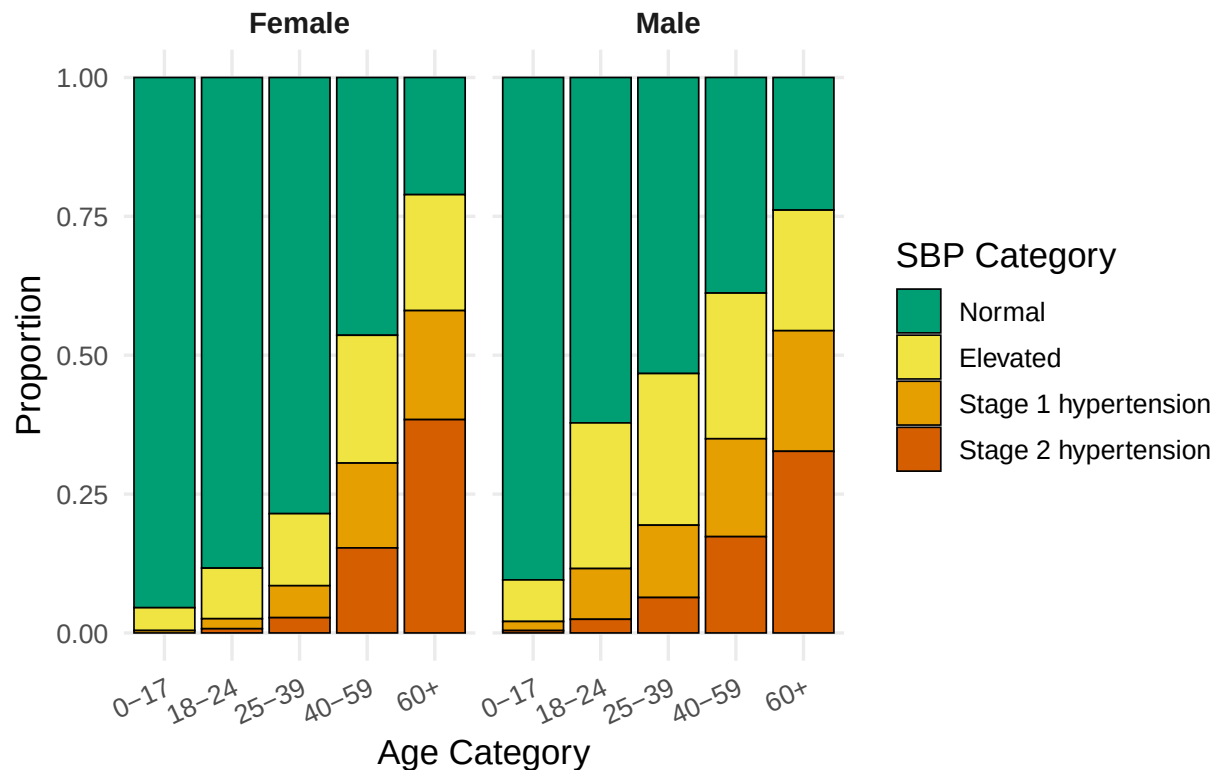
```
## [1] "V1"          "patient_id"    "bp_comment"    "systolic_bp_1"
## [5] "diastolic_bp_1" "systolic_bp_2" "diastolic_bp_2" "systolic_bp_3"
## [9] "diastolic_bp_3" "systolic_bp_4" "diastolic_bp_4" "wave_id"
## [13] "gender"        "age"           "ethnicity_2"   "education"
## [17] "marital_status" "family_size"   "family_income" "age_cat"
## [21] "avg_sbp"       "sbp_category"
```

```

##the new data set (nhanes_clean) has 23 variables and 21,604 observations
##using facet_wrap() to investigate hypertension varies across age and gender
ggplot(nhanes_clean, aes(x = age_cat, fill = sbp_category)) +
  geom_bar(position = "fill", color = "black", linewidth = 0.3) +
  ##using the facet_wrap to display the relationship
  facet_wrap(~ gender) +
  scale_fill_manual(
    values = c(
      "Normal" = "#009E73", # Green
      "Elevated" = "#F0E442", # Yellow
      "Stage 1 hypertension" = "#E69F00", # Orange
      "Stage 2 hypertension" = "#D55E00" # Red-Orange
    )
  ) +
  labs(
    title = "Distribution of Blood Pressure Categories by Age and Gender",
    x = "Age Category",
    y = "Proportion",
    fill = "SBP Category"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(color = "gray30", size = 11),
    axis.text.x = element_text(angle = 25, hjust = 1),
    legend.position = "right",
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold", size = 11)
  )

```

Distribution of Blood Pressure Categories by Age and Gender



```
##use of 4 plots to describe the final sample size
old_dataset <- nrow(nhanes_data)
new_dataset <- nrow(nhanes_clean)
n_removed <- old_dataset - new_dataset
percent_removed <- 100 * n_removed / old_dataset
```

```
sample_sizes <- data.frame(
  Dataset = c("old sample", "new sample"),
  Observations = c(old_dataset, new_dataset)
)
```

```
sample_sizes
```

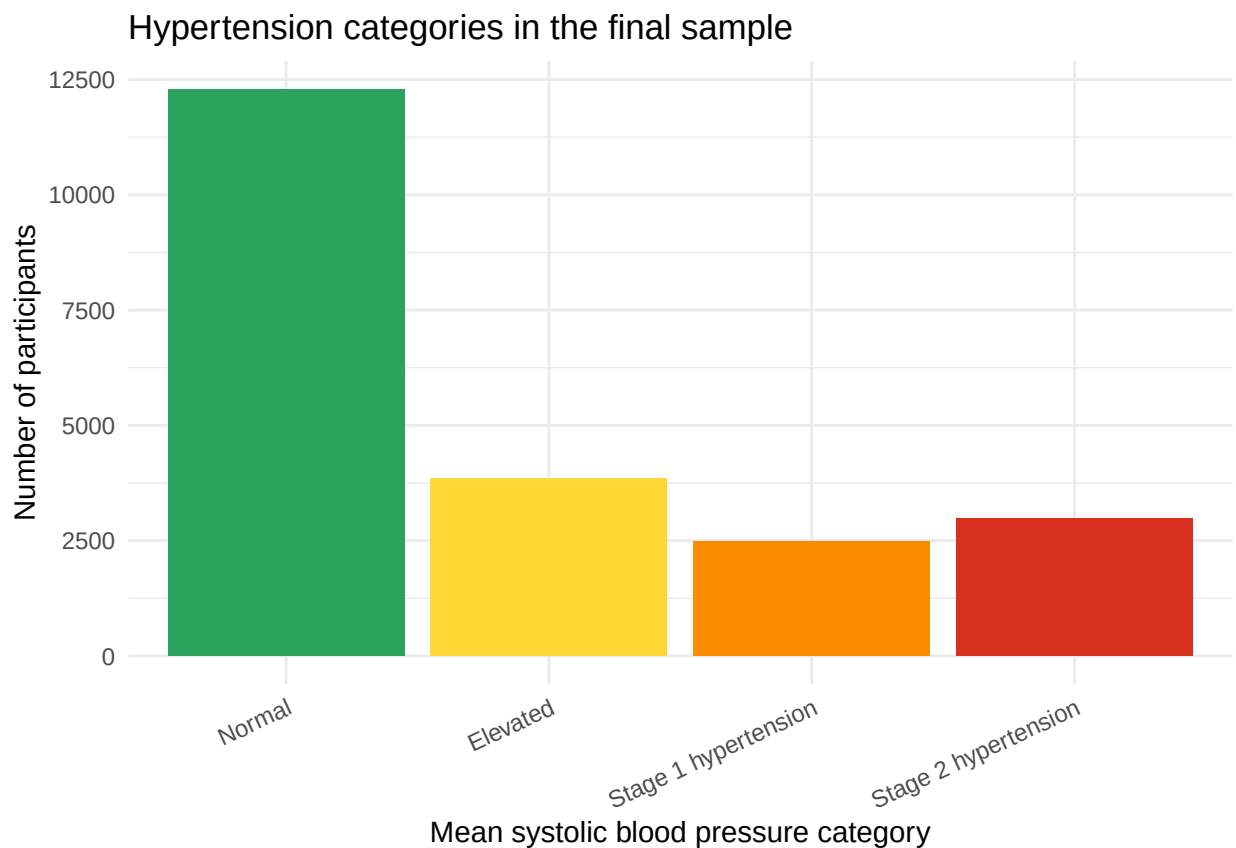
```
##      Dataset Observations
## 1 old sample      29400
## 2 new sample      21604
```

```
##Plot showing hypertension categories
##defining
sbp_colors <- c(
  "Normal" = "#2CA25F",
  "Elevated" = "#FDD835",
  "Stage 1 hypertension" = "#FB8C00",
  "Stage 2 hypertension" = "#D7301F"
)
```

```

plot_sbp <- ggplot(
  nhanes_clean,
  aes(x = sbp_category, fill = sbp_category)
) +
  geom_bar() +
  scale_fill_manual(values = sbp_colors, drop = FALSE) +
  labs(
    title = "Hypertension categories in the final sample",
    x = "Mean systolic blood pressure category",
    y = "Number of participants"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 25, hjust = 1)
  )
plot_sbp

```



```

##Plot of gender distribution
p_gender <- ggplot(nhanes_clean, aes(x = gender, fill = gender)) +
  geom_bar() +
  labs(
    title = "Gender distribution in the final sample",
    x = "Gender",

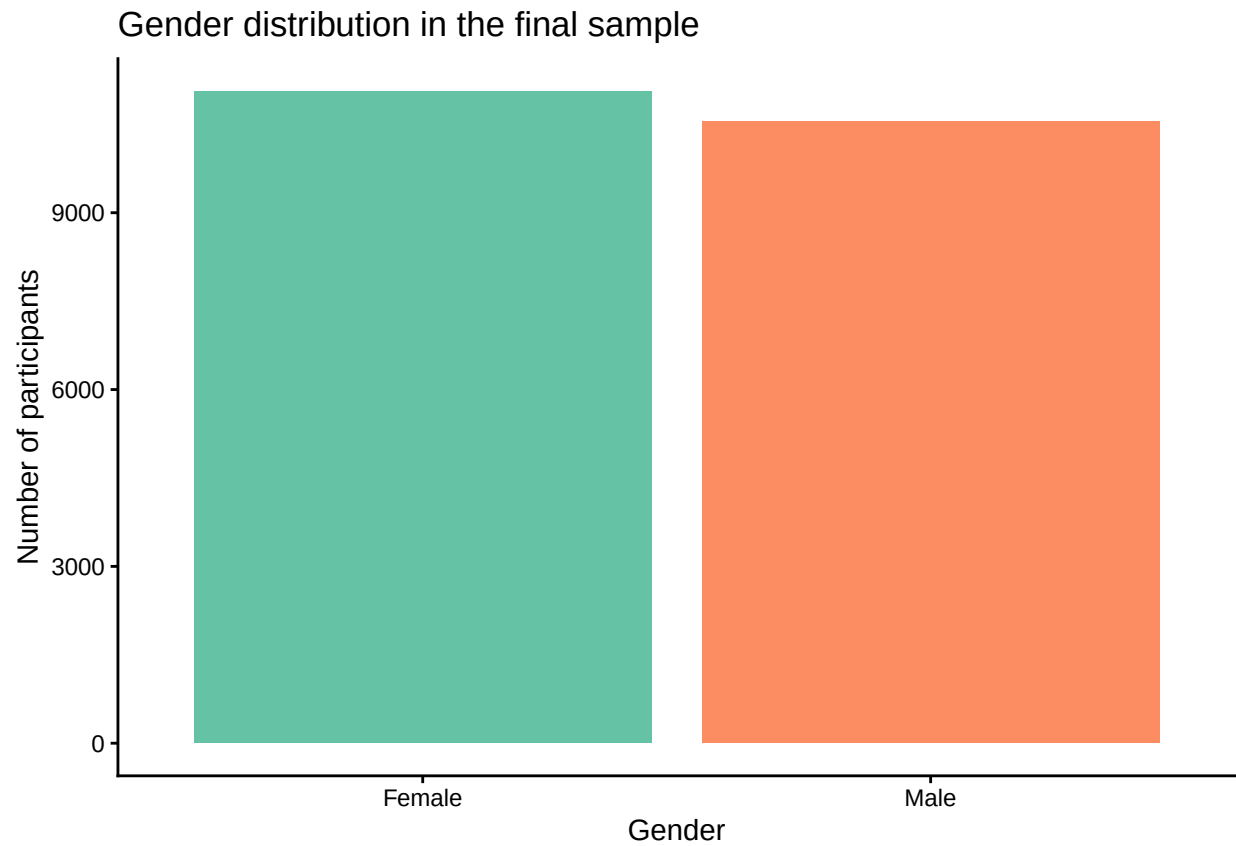
```

```

  y = "Number of participants"
) +
scale_fill_brewer(palette = "Set2") +
theme_classic() +
theme(legend.position = "none")

```

p_gender



```

##reuse one saved plot from Exercise 2 and compare it to the same variable's distribution in your final
##sample
##plot from exercise 2 was saved as "age_distribution_plot.png"
library(dplyr)
library(ggplot2)

# Recreate the data used for the original saved plot
age_comparison <- bind_rows(
  nhanes_data %>%
    filter(!is.na(age)) %>%
    transmute(
      age,
      sample = "Original sample"
    ),

  nhanes_clean %>%
    filter(!is.na(age)) %>%

```

```

    transmute(
      age,
      sample = "Final sample"
    )
  )

# Faceted comparison using the same histogram style
age_comparison <- bind_rows(
  nhanes_data %>%
    filter(!is.na(age)) %>%
    select(age) %>%
    mutate(sample = "Original sample"),

  nhanes_clean %>%
    filter(!is.na(age)) %>%
    select(age) %>%
    mutate(sample = "Final sample")
)

names(age_comparison)

```

```
## [1] "age"      "sample"
```

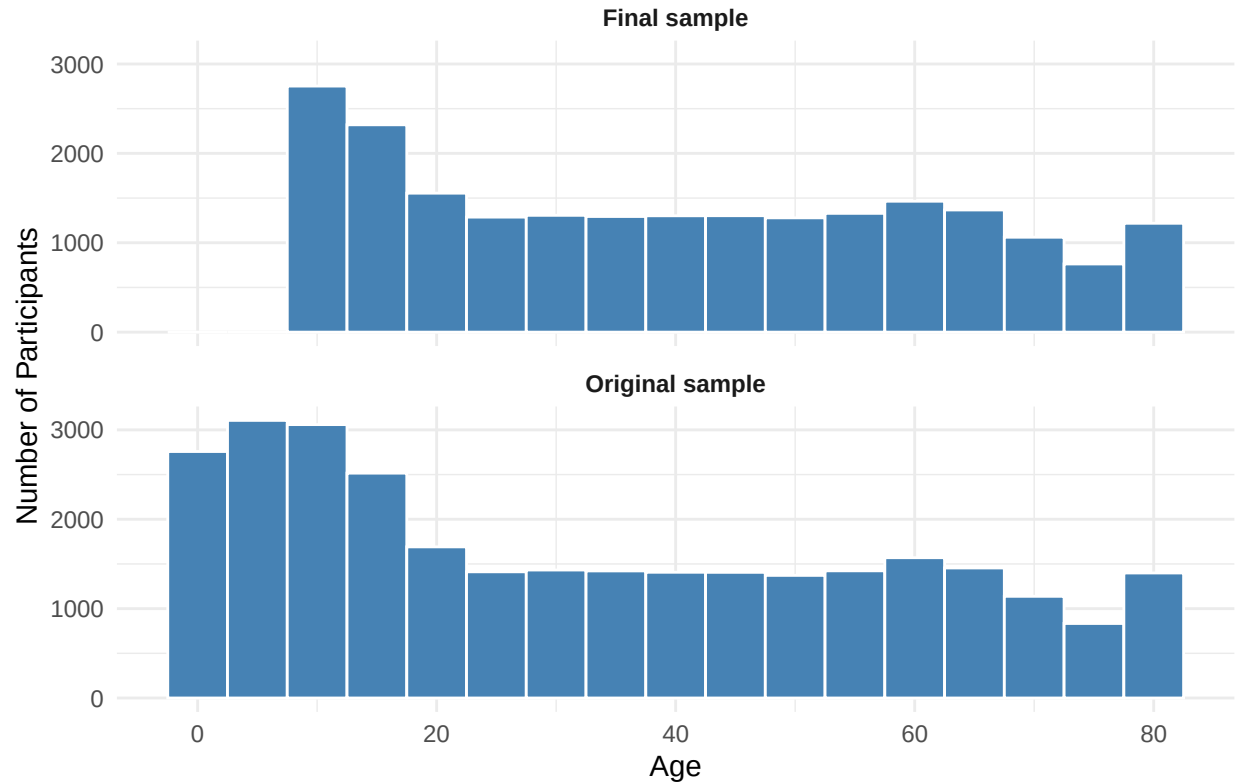
```

##the faceted plot
age_comparison_plot2 <- ggplot(
  age_comparison,
  aes(x = age)
) +
  geom_histogram(
    binwidth = 5,
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ sample, ncol = 1) +
  labs(
    title = "Age Distribution original and final distribution",
    x = "Age",
    y = "Number of Participants"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold"),
    strip.text = element_text(face = "bold")
  )

age_comparison_plot2

```

Age Distribution original and final distribution



##Comparison of the plots before and after. The age distribution generally remained similar before and after. However, the original histogram show more concentration in the early bins and the final shows more flatter bins. This means that those removed because of missing gender, mean SBP may not have been evenly distributed. In conclusion, filtering may have reduced the sample size but it doesnt change the distribution of age.

A short report

To analyze systolic blood pressure (SBP), i calculated an average for each participant using up to four available readings. Participants missing all four readings were excluded, along with anyone missing age or gender data.

The final sample contains fewer participants than the original data set, and its age distribution is not identical to the original distribution. Although the overall age range remains similar, the final sample is flatter at the younger end, where the original data set shows more pronounced bars. This suggests that missing data were not distributed entirely at random across age groups and that filtering may have changed the representation of younger participants.

Younger participants are more commonly classified as having normal or elevated blood pressure, whereas older participants show a greater proportion of Stage 1 and Stage 2 hypertension. These age-related patterns are visible in both genders, although the proportions may differ somewhat between males and females within particular age groups.

Overall, the visualizations suggest an association between increasing age and higher SBP in this cohort.